郝宇

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我是华为鸿蒙系统安全组的系统安全研究员，负责鸿蒙操作系统的内存安全工作。我的研究聚焦操作系统安全与 C/C++ 程序分析，结合 Fuzzing、静态分析、符号执行和大语言模型推理，保障 Linux、Android、鸿蒙等真实内核的安全。代表性工作包括内核 Fuzzing 规格自动生成（SyzDescribe，S&P 23；SyzSpec，CCS 25 杰出论文奖）和 LLM 辅助静态分析（E&V；OOPSLA 24）。研究成果发表于 S&P、USENIX Security、CCS、NDSS、ICSE、FSE、OOPSLA 和 TSE 等会议/期刊，并产出多个开源工具、受邀报告以及 Linux 内核漏洞报告和补丁。

教育经历

09/2018 – 09/2024 **博士**，加州大学河滨分校，计算机科学
导师：Zhiyun Qian

09/2015 – 06/2018 **硕士**，西安交通大学，控制科学与工程
导师：刘烃

09/2011 – 06/2015 **学士**，西安交通大学，控制科学与工程

09/2008 – 06/2011 呼和浩特第二中学，内蒙古自治区

工作经历

10/2024 – 至今 **系统安全研究员**
鸿蒙系统安全组，华为，中国
获得华为“天才少年”录用
鸿蒙系统安全、Fuzzing、LLM 辅助安全分析、静态分析
- 负责鸿蒙内核 Fuzzing 技术路线，面向内核及系统组件的内存安全漏洞挖掘。
- 负责鸿蒙操作系统 Rust 技术路线，在新系统组件中从根源上消除内存安全类漏洞。
- 推动 AI for Security，将大语言模型应用于安全分析。

06/2023 – 09/2023 **研究实习生**，导师：Weidong Cui
系统安全与隐私研究组，微软研究院，美国
LLM 辅助安全分析与程序分析
- 提出 E&V，较早探索让大语言模型通过执行伪代码并验证结果来完成静态分析，在不依赖额外工具的情况下降低幻觉。
- 将 E&V 用于 Linux 内核崩溃分诊（定位应修补的函数），在七类漏洞的 170 个崩溃上准确率超过 80%。

06/2022 – 09/2022 **研究实习生**，导师：Kris Alder
Android 安全团队，谷歌，美国
Linux/Android 内核安全、Fuzzing、静态分析
- 使用静态分析增强 Android 内核 Fuzzing（内部工具）。

09/2018 – 09/2024 **研究生研究员**，导师：Zhiyun Qian
加州大学河滨分校，网络安全与隐私研究和教育中心（CRESP）
- 构建了下列课题组内核安全项目中的欠约束符号执行及大部分静态分析工具。
- 内核 Fuzzing：揭示内核 Fuzzing “依赖难题”背后的根本原因（ICSE 22，Google Research Paper Rewards）；先后基于静态分析（SyzDescribe，S&P 23）和欠约束符号执行（SyzSpec，CCS 25，杰出论文奖）实现系统调用规格自动生成；为驱动 Fuzzing 的依赖推断提供思路（SyzGen++，S&P 24）。
- 漏洞分诊与可利用性：参与设计并实现 Fuzzing 所发现漏洞精确二分定位中的符号执行（SymBisect，USENIX Security 24）；实现 Linux 生态漏洞可利用性评估中的静态分析（SyzBridge，NDSS 24）。
- 精确静态分析：实现 Linux 内核未初始化使用漏洞检测中的欠约束符号执行，消除约 87% 的误报（UBITect，FSE 20），并支持增量检测（NDSS 22）。
- LLM 辅助程序分析：为 LLM 与静态分析结合的漏洞检测提供思路并实现其中的静态分析（FSE 23，OOPSLA 24）。

06/2013 – 06/2018 **研究助理**，导师：刘烃
西安交通大学，智能网络与网络安全教育部重点实验室
并发安全、符号执行

毕业论文

[博士] Expanding the Boundaries of OS Kernel Fuzzing Permalink
- [C9, ICSE 22] [C10, S&P 23] [C17, CCS 25, Distinguished Paper Award]

[硕士] 基于并行程序不确定时序交织的恶意代码隐藏方法研究
- [C3, QRS 21, Best Paper Award] [J3, *IEEE Transactions on Reliability*]

出版论文

[C]: 会议, [J]: 期刊, [P]: 预印本
小计: CCF-A 类会议 12 篇 (一作 3 篇, 二作 6 篇), CCF-A 类期刊 1 篇, 正式发表论文 21 篇

[C18] ISSRE 26
NeuroMerge: ML-Guided State Merging for Efficient Symbolic Execution
Shenghan Zheng, Shitong Zhu, **Yu Hao**, Xingyu Li, Keyu Man, Zheng Zhang, Qing Deng, Zhiyun Qian, Srikanth Krishnamurthy
IEEE International Symposium on Software Reliability Engineering, ISSRE 26.

[C17] CCS 25
SyzSpec: Specification Generation for Linux Kernel Fuzzing via Under-Constrained Symbolic Execution
Yu Hao, Juefei Pu, Xingyu Li, Zhiyun Qian, Ardalan Amiri Sani
ACM SIGSAC Conference on Computer and Communications Security, CCS 25. Tool
- [Distinguished Paper Award]

[C16] S&P 25
SCAD: Towards a Universal and Automated Network Side-Channel Vulnerability Detection
Keyu Man, Zhongjie Wang, **Yu Hao**, Shenghan Zheng, Xin'an Zhou, Yue Cao, Zhiyun Qian
IEEE Symposium on Security and Privacy, S&P 25. Tool

[C15] USENIX Security 24
SymBisect: Accurate Bisection for Fuzzer-Exposed Vulnerabilities
Zheng Zhang, **Yu Hao**, Weiteng Chen, Xiaochen Zou, Xingyu Li, Haonan Li, Yizhuo Zhai, Zhiyun Qian, Billy Lau
USENIX Security Symposium 2024. Tool
- [Linux Security Summit 25]

[C14] OOPSLA 24
Enhancing Static Analysis for Practical Bug Detection: An LLM-Integrated Approach
Haonan Li, **Yu Hao**, Yizhuo Zhai, Zhiyun Qian
ACM SIGPLAN International Conference on Object-Oriented Programming Systems, Languages, and Applications, OOPSLA 24. Tool

[C13] S&P 24
SyzGen++: Dependency Inference for Augmenting Kernel Driver Fuzzing
Weiteng Chen, **Yu Hao**, Zheng Zhang, Xiaochen Zou, Dhilung Kirat, Shachee Mishra, Douglas Schales, Jiyong Jang, Zhiyun Qian
IEEE Symposium on Security and Privacy, S&P 24. Tool

[C12] NDSS 24
SyzBridge: Bridging the Gap in Exploitability Assessment of Linux Kernel Bugs in the Linux Ecosystem
Xiaochen Zou, **Yu Hao**, Zheng Zhang, Juefei Pu, Weiteng Chen, Zhiyun Qian
Network and Distributed System Security Symposium, NDSS 24. Tool

[P1]
E&V: Prompting Large Language Models to Perform Static Analysis by Pseudo-code Execution and Verification
Yu Hao, Weiteng Chen, Ziqiao Zhou, Weidong Cui
- [AGI Leap Summit 2024] [Symposium on the Science of Security 24]

[C11] FSE 23 IVR
Assisting Static Analysis with Large Language Models: A ChatGPT Experiment
Haonan Li, **Yu Hao**, Yizhuo Zhai, Zhiyun Qian
The ACM International Conference on the Foundations of Software Engineering, Ideas, Visions and Reflections, FSE 23 IVR. Tool

[C10] S&P 23
SyzDescribe: Principled, Automated, Static Generation of Syscall Descriptions for Kernel Drivers
Yu Hao, Guoren Li, Xiaochen Zou, Weiteng Chen, Shitong Zhu, Zhiyun Qian, Ardalan Amiri Sani
IEEE Symposium on Security and Privacy, S&P 23. Tool Result
- [Linux Security Summit 23] [Qualcomm Product Security Summit 23]
- [Symposium on the Science of Security 24]

[C9] ICSE 22
Demystifying the Dependency Challenge in Kernel Fuzzing
Yu Hao, Hang Zhang, Guoren Li, Xingyun Du, Zhiyun Qian, Ardalan Amiri Sani
IEEE/ACM International Conference on Software Engineering, ICSE 22. Tool
- [Google Research Paper Rewards]

[C8] **NDSS 22**
Progressive Scrutiny: Incremental Detection of UBI bugs in the Linux Kernel
Yizhuo Zhai, **Yu Hao**, Zheng Zhang, Weiteng Chen, Guoren Li, Zhiyun Qian, Chengyu Song, Manu Sridharan, Srikanth V. Krishnamurthy, Trent Jaeger, Paul Yu
Network and Distributed System Security Symposium, NDSS 22. Tool
- [2023 Cyber Security CRA Capstone Poster]

[C7] ACSAC 21
Eluding ML-based Adblockers With Actionable Adversarial Examples
Shitong Zhu, Zhongjie Wang, Xun Chen, Shasha Li, Keyu Man, Umar Iqbal, Zhiyun Qian, Kevin S Chan, Srikanth V Krishnamurthy, Zubair Shafiq, **Yu Hao**, Guoren Li, Zheng Zhang, Xiaochen Zou
Annual Computer Security Applications Conference, ACSAC 21. Tool

[C6] **CCS 21**
Themis: Ambiguity-Aware Network Intrusion Detection based on Symbolic Model Comparison
Zhongjie Wang, Shitong Zhu, Keyu Man, Pengxiong Zhu, **Yu Hao**, Zhiyun Qian, Srikanth V. Krishnamurthy, Tom La Porta, Michael J. De Lucia
ACM SIGSAC Conference on Computer and Communications Security, CCS 21. Tool

[C5] **CCS 21**
Statically Discovering High-Order Taint Style Vulnerabilities in OS Kernels
Hang Zhang, Weiteng Chen, **Yu Hao**, Guoren Li, Yizhuo Zhai, Xiaochen Zou, Zhiyun Qian
ACM SIGSAC Conference on Computer and Communications Security, CCS 21. Tool

[C4] **FSE 20**
UBITect: A Precise and Scalable Method to Detect Use-before-Initialization Bugs in Linux Kernel
Yizhuo Zhai, **Yu Hao**, Hang Zhang, Daimeng Wang, Chengyu Song, Zhiyun Qian, Mohsen Lesani, Srikanth V. Krishnamurthy, Paul Yu
ACM SIGSOFT International Symposium on Foundations of Software Engineering, FSE 20. Tool
- [2023 Cyber Security CRA Capstone Poster]

[J3]
ConcSpectre: Be Aware of Forthcoming Malware Hidden in Concurrent Programs
Yang Liu, Ming Fan, Ting Liu, **Yu Hao**, Zisen Xu, Kai Chen, Hao Chen, and Yan Cai
IEEE Transactions on Reliability. Code Result

[C3] QRS 21
ConcSpectre: Be Aware of Forthcoming Malware Hidden in Concurrent Programs
Yang Liu, Ming Fan, Ting Liu, **Yu Hao**, Zisen Xu, Kai Chen, Hao Chen, and Yan Cai
IEEE International Conference on Software Quality, Reliability, and Security, QRS 21. Code Result
- [Best Paper Award]

[J2] **TSE**
Tell You a Definite Answer: Whether Your Data is Tainted During Thread Scheduling
Xiaodong Zhang, Zijiang Yang, Qinghua Zheng, **Yu Hao**, Pei Liu, Ting Liu
*IEEE Transactions on Software Engineering, **TSE**.* Tool Benchmarks Result
- [S&P 17 Poster] Patent: [PCT] [CN]

[J1]
Debugging Multithreaded Programs as if They Were Sequential
Xiaodong Zhang, Zijiang Yang, Qinghua Zheng, **Yu Hao**, Pei Liu, Lechen Yu, Ting Liu
IEEE Access. Tool

[C2] ICST 17
Automated Testing of Definition-Use Data Flow for Multithreaded Programs
Xiaodong Zhang, Zijiang Yang, Qinghua Zheng, Pei Liu, Jialiang Chang, **Yu Hao**, Ting Liu
IEEE International Conference on Software Testing, Verification and Validation, ICST 17. Tool

[C1] SATE 17
Debugging Multithreaded Programs as if They Were Sequential
Xiaodong Zhang, Zijiang Yang, Qinghua Zheng, **Yu Hao**, Pei Liu, Lechen Yu, Ming Fan, Ting Liu
IEEE International Conference on Software Analysis, Testing and Evolution, SATE 16. Tool

专利

WO2017181628 - TAINT ANALYSIS METHOD EMPLOYING SYMBOLIC COMPUTATION AND USED FOR DYNAMIC PARALLEL PROGRAM [PCT] [CN]

演讲，报告

- E&V: Prompting Large Language Models to Perform Static Analysis by Pseudo-code Execution and Verification
  - *AGI Leap Summit 2024* Website
  - *Symposium on the Science of Security, HoTSoS 2024* Website
- SyzDescribe: Principled, Automated, Static Generation of Syscall Descriptions for Kernel Drivers
  - *Linux Security Summit North America 2023* Website Slides Recording
  - *Qualcomm Product Security Summit 2023* Website
  - *44th IEEE Symposium on Security and Privacy, S&P 2023* Website Recording
  - *Symposium on the Science of Security, HoTSoS 2024* Website

● Demystifying the Dependency Challenge in Kernel Fuzzing
- *44th IEEE/ACM International Conference on Software Engineering, ICSE 2022* Website Slides Recording

荣誉, 奖励

2025	CCS 2025, Distinguished Paper Award
2024	Laxmi Bhuyan Fellowship Award, University of California, Riverside
	Dissertation Completion Fellowship Award, University of California, Riverside
2023	IEEE S&P Student Travel Grant
	Google Research Paper Rewards
	The Linux Foundation's Travel Fund
2021	QRS 2021 Best Paper Award
2018	Dean's Distinguished Fellowship, University of California, Riverside
	西安交通大学优秀硕士毕业生
2013	西安交通大学程序设计竞赛，二等奖
	西安交通大学数学建模竞赛，二等奖
2010	内蒙古高中数学联赛，一等奖
	内蒙古高中物理联赛，三等奖
	内蒙古高中化学联赛，三等奖
2007	全国奥林匹克数学竞赛，金牌

学术服务

程序委员会委员	CCS 2027, 2026 RAID 2026 FORGE@ICSE 2025, 2024 MSR 2024 (Junior PC)
审稿人	TIFS 2026, 2024, 2023 TDSC 2026 TOSEM 2026, 2025 Computing Surveys 2026 EAI SecureComm 2023
外部审稿人	S&P 2025, 2024, 2021, 2020 CCS 2024 USENIX Security 2025, 2021 NDSS 2025, 2021, 2020
AEC	EuroSys 2023 ISSTA 2024 ECOOP 2024

推荐人

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实习导师 微软研究院	**Weidong Cui** *Partner Research Manager* 系统安全与隐私研究组，微软研究院，美国 联系方式: Weidong.Cui@microsoft.com

硕士导师 **刘烃** 教授，副院长
西安交通大学，网络空间安全学院
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